RAJALAKSHMI ENGINEERING COLLEGE
Approved by AICTE | Affiliated to Anna University | Accredited by NAAC

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence
V Semester B. Tech AIDS

AD23532 – Principles of Data Science

Ex2: Data Discovery and Preparation

Aim: To explore the Telco Customer Churn dataset, find outliers using the IQR method, study the factors affecting churn and prepare the data for modelling.

Learning Objective:
The objective of this experiment is to perform data discovery and exploratory data analysis on a real-world dataset.

Question:
A telecommunications company has observed an increase in customer churn, leading to revenue loss. The management wants to understand customer behavior and identify patterns before developing a machine learning model to predict churn.

As a Data Analyst, your responsibility is to discover meaningful insights from the dataset, identify data quality issues, perform exploratory data analysis, detect outliers, and prepare the dataset for predictive analytics.

Tasks to Perform:

1. Reuse Ex:1 Steps 1- 6, 10.
2. Analyze the distribution of: Customer Churn, Gender, Contract Type, Payment Method.
3. Visualize the distribution of: MonthlyCharges, TotalCharges Tenure using Histograms and Boxplots.
4. Detect outliers in numerical features using the Interquartile Range (IQR) method.
5. Create suitable visualizations (Count Plot, Box Plot, Histogram, or Scatter Plot) to study the relationship between: Contract Type and Churn, Internet Service and Churn, Monthly Charges and Churn, Tenure and Churn.
6. Do the below analytical questions:
   - Which contract type has the highest customer churn?
   - Which payment method is most frequently used by customers who churn?
   - Do customers with shorter tenure tend to churn more often?
   - Which customer group has the highest monthly charges?
   - Which Internet Service category experiences the highest churn?
   - Is there any relationship between monthly charges and churn?
   - Which features appear to be the strongest indicators of customer churn?
7. Split the cleaned dataset into Training (80%) and Testing (20%) datasets.
8. Save the final processed dataset as a new CSV file named CustomerChurn_Preprocessed.csv.

Solution:

```
# Reused from Ex1 (import, load, explore, clean)
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```



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```
df = df.drop_duplicates()
df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())

# Distribution of Churn, Gender, Contract, PaymentMethod
fig, ax = plt.subplots(1, 4, figsize=(20, 4))
for a, c in zip(ax, ["Churn", "gender", "Contract", "PaymentMethod"]):
    sns.countplot(data=df, x=c, ax=a)
    a.tick_params(axis="x", rotation=25)
plt.tight_layout()
plt.show()
for c in ["Churn", "gender", "Contract", "PaymentMethod"]:
    print((df[c].value_counts(normalize=True) * 100).round(1))

# Histograms and boxplots
num = ["MonthlyCharges", "TotalCharges", "tenure"]
fig, ax = plt.subplots(2, 3, figsize=(15, 7))
for i, c in enumerate(num):
    sns.histplot(df[c], kde=True, ax=ax[0, i])
    sns.boxplot(x=df[c], ax=ax[1, i])
plt.tight_layout()
plt.show()

# Outlier detection using IQR
for c in num:
    q1, q3 = df[c].quantile([0.25, 0.75])
    iqr = q3 - q1
    out = df[(df[c] < q1 - 1.5 * iqr) | (df[c] > q3 + 1.5 * iqr)]
    print(f"{c}: {len(out)} outliers")

# Relationship with churn
fig, ax = plt.subplots(2, 2, figsize=(12, 9))
sns.countplot(data=df, x="Contract", hue="Churn", ax=ax[0, 0])
sns.countplot(data=df, x="InternetService", hue="Churn", ax=ax[0, 1])
sns.boxplot(data=df, x="Churn", y="MonthlyCharges", ax=ax[1, 0])
sns.boxplot(data=df, x="Churn", y="tenure", ax=ax[1, 1])
plt.tight_layout()
plt.show()

# Analytical questions
yes = lambda s: (s == "Yes").mean() * 100
print(df.groupby("Contract")["Churn"].apply(yes).round(1))              # Q1
print(df[df.Churn == "Yes"]["PaymentMethod"].value_counts())            # Q2
print(df.groupby("Churn")[["tenure", "MonthlyCharges"]].mean().round(1)) # Q3,Q6
print(df.groupby("InternetService")["MonthlyCharges"].mean().round(1))   # Q4
print(df.groupby("InternetService")["Churn"].apply(yes).round(1))        # Q5
tmp = df.drop(columns="customerID").copy()
for c in tmp.select_dtypes(exclude=np.number):
    tmp[c] = LabelEncoder().fit_transform(tmp[c])
print(tmp.corr()["Churn"].abs().sort_values(ascending=False).head(6).round(3))  # Q7

# Encode, scale, split and save
df = df.drop(columns="customerID")
for c in ["gender", "Partner", "Dependents", "Churn", "PhoneService",
          "PaperlessBilling"]:
    df[c] = LabelEncoder().fit_transform(df[c])
df = pd.get_dummies(df, dtype=int)
df[num] = StandardScaler().fit_transform(df[num])

X, y = df.drop(columns="Churn"), df["Churn"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
print("Train:", X_train.shape, "Test:", X_test.shape)
df.to_csv("CustomerChurn_Preprocessed.csv", index=False)
```

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Answers:

1. Month-to-month contracts have the highest churn (42.7%, against 11.3% for one year and 2.8% for two year).

2. Electronic check is the most used payment method among churned customers (1,071 of 1,869).

3. Yes. Churned customers have an average tenure of about 18 months, while other customers have about 37.6 months.

4. Fiber optic customers have the highest monthly charges (average about 91.5).

5. Fiber optic has the highest churn (41.9%, against 19.0% for DSL and 7.4% for no internet).

6. Yes. Churned customers pay more on average (74.4) than the others (61.3).

7. Contract (0.40), tenure (0.35), OnlineSecurity (0.29), TechSupport (0.28) and TotalCharges (0.20) have the highest correlation with churn.

No outliers were found in MonthlyCharges, TotalCharges or tenure. The overall churn rate is 26.5%.

Result:

The data was explored and the questions were answered. It was split into 5,634 training and 1,409 testing records and saved as CustomerChurn_Preprocessed.csv.

Ex3a: Model Planning and Building - Classification

Learning Objective:
The objective of this experiment is to enable students to plan, build, and evaluate basic machine learning models using real-world datasets.

Aim: To build Logistic Regression, Decision Tree and Random Forest models to predict customer churn and compare their performance.

Problem Statement:
A telecommunications company has noticed that a growing number of customers are discontinuing its services, resulting in reduced revenue and increased customer acquisition costs. After completing data preprocessing and exploratory analysis, the company now aims to develop predictive models that can identify customers who are likely to churn. Early prediction of customer churn will enable the company to implement targeted retention strategies, improve customer satisfaction, and reduce business losses.

As a Data Scientist, your task is to build and compare multiple classification models to predict whether a customer will churn. You will train different machine learning algorithms, evaluate their performance using appropriate classification metrics, and identify the most effective model for customer churn prediction. Finally, interpret the important factors influencing customer churn and communicate your findings through suitable visualizations and performance comparisons.

Tasks to Perform:

1. Use CustomerChurn_Preprocessed.csv file which you created in Ex2.
2. Split the cleaned dataset into Training (80%) and Testing (20%) datasets.

Model Building

3. Train a Logistic Regression classifier and predict the customer churn on the test dataset.
4. Train a Decision Tree Classifier and predict the customer churn on the test dataset.
5. Train a Random Forest Classifier and predict the customer churn on the test dataset.

Model Evaluation

6. Evaluate each model using Confusion Matrix, Accuracy, Precision, Recall, F1-Score.
7. Compare the performance of all three models using a single bar chart showing the evaluation metrics.
8. Identify the best-performing model based on the evaluation results.

Business Recommendation

9. Based on the best-performing model, provide three recommendations that the telecommunications company can implement to reduce customer churn.

Solution:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
```